

Mikhail Volkov

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Born: 30.04.1996 Penza, Russia



Postdoctoral researcher in chemoinformatics

Education

- 2013–2019 **Department of Chemistry, Specialist degree in Fundamental and Applied Chemistry**, *Lomonosov Moscow State University*, Moscow, Russia.
- 2019–2023 **École doctorale des Sciences Chimiques – ED222**, *University of Strasbourg*, Strasbourg, France.

Professional experience

- June–July 2018 **Laboratory of quality control**, *Public Joint Stock Company “Biosintez” (Sun Pharmaceutical Industries Ltd.)*, Penza, Russia.
Internship – Laboratory assistant
- 2017–2019 **Laboratory of medicinal chemistry**, *Faculty of Chemistry, Lomonosov Moscow State University*, Moscow, Russia.
Graduate thesis
- November 2019–April 2023 **Laboratoire d’innovation thérapeutique – UMR 7200**, *Faculty of Pharmacy, University of Strasbourg*, Illkirch, France.
PhD Student
- August 2023–Present **Laboratory of Chemoinformatics – UMR 7140**, *University of Strasbourg / CNRS*, Strasbourg, France.
Postdoctoral researcher

Scientific interests

Medicinal chemistry, machine learning in chemistry, QSAR/QSPR, deep generative models, graph neural networks

Publications

- May 2022 **On the Frustration to Predict Binding Affinities from Protein–Ligand Structures with Deep Neural Networks**, *Mikhail Volkov, Joseph-André Turk, Nicolas Drizard, Nicolas Martin, Brice Hoffmann, Yann Gaston-Mathé, and Didier Rognan*, *Journal of Medicinal Chemistry*.
Journal Article

Teaching experience

- November 2018–March 2019 **Molecular modeling in drug design course**, *Sechenov First Moscow State Medical University*, Moscow, Russia.
Professor’s assistant at seminars

Participation in conferences

- April 2017 **Analysis of small molecules binding modes with dengue virus envelope protein trimer**, *XXIV Russian National Congress “Man and Drug”, section of Bioinformatics and Computer-Aided Drug Design*, Moscow, Russia.
Oral presentation

- July 2017 **Analysis of small molecules binding with dengue virus envelope protein trimer**, *3rd Kazan Summer School on Chemoinformatics*, Kazan, Russia.
Poster presentation
- April 2018 **Modeling of small molecules binding with dengue virus envelope protein trimer**, *Current issues of epidemiology, diagnostics, treatment and prevention of infectious and oncology diseases*, Moscow, Russia.
Oral presentation
- September 2021 **Applicability of graph neural networks to predict binding affinities from protein-ligand structures**, *Congrès GGMM SFCi 2021*, Lille, France.
Poster presentation
- May 2022 **Applicability of graph neural networks to binding affinities prediction from protein-ligand structures**, *17th German Conference on Cheminformatics*, Garmisch-Partenkirchen, Germany.
Poster presentation

Awards

- April 2017 **3rd place in young scientists' awards**, *XXIV Russian National Congress "Man and Drug"*, section of Bioinformatics and Computer-Aided Drug Design, Moscow, Russia.
- July 2017 **Prize for the best young scientist's poster presentation**, *3rd Kazan Summer School on Chemoinformatics*, Kazan, Russia.
- January 2018 **3rd place in hackathon**, *3rd Advanced in silico Drug Design workshop/hackathon 2018*, Olomouc, Czech Republic.

Technical skills

- Programming languages Solid experience with: **Python** (package development, task automation scripts, data analysis in Jupyter), **C++** (development of CLI tools, ML inference), **SQL** (database development with SQLite3, queries in ML dataloading pipelines), **Bash** (task automation scripts in Linux)
- ML libraries and frameworks Scikit-Learn, Pytorch, Tensorflow, ONNX, DGL (development of Graph NN architectures), Pytorch-Lightning (integration of metrics, parameter tracking)
- Chemoinformatics libraries RDkit, OpenEye Toolkits, Indigo

Soft skills

- Languages **English** — IELTS 7.5 (May 2019) – 3 years of working in an English speaking team. **German** – B1-B2 – DSH 2 certificate (May 2016). **French** – A2-B1
- Communication skills Scientific communication, scientific writing skills
- Other Team working skills, Remote working experience, Research intern supervision

Other interests

- Music Membership in an orchestra of breton music